

ROHAN S. BEDAGE

Pune, MH | +91 8329234445 | rohansbedage@gmail.com | linkedin.com/in/rohanbedage | github.com/rohansbedage

SUMMARY

Research-oriented AI Engineer focused on Generative AI, Large Language Models, Agentic AI, Computer Vision, and intelligent systems, with hands-on experience designing advanced deep learning architectures, autonomous AI frameworks, and synthetic data generation pipelines.

EDUCATION

Defence Institute of Advanced Technology (DIAT), Pune <i>M.Tech - Artificial Intelligence CGPA: 7.8/10</i>	Pune, MH, IN <i>July 2024 – July 2026</i>
P.E.S Modern College of Engineering, Pune <i>B.E - Artificial Intelligence & Machine Learning CGPA: 8.46/10</i>	Pune, MH, IN <i>Dec 2021 – June 2024</i>
Cusrow Wadia Institute of Technology, Pune <i>Diploma – Computer Science & Engineering 93.53%</i>	Pune, MH, IN <i>July 2019 – Oct 2021</i>

TECHNICAL SKILLS

Languages: Python, C++, SQL, PHP, HTML, CSS, Bootstrap
AI / ML Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NLTK
Generative & Agentic AI: Ollama, FastAPI, LLM Orchestration, Multi-Agent Systems
Computer Vision & Signals: VMamba, GANs, CNNs, FFT, Wavelet Transform, Spectral Processing
NLP & LLMs: BERT, RoBERTa, DistilBERT, Hugging Face Transformers, Knowledge Graphs
Tools & Platforms: AWS, Docker, Git, MATLAB, Linux
Domains: Generative AI, Agentic AI, Computer Vision, NLP, Healthcare AI, Synthetic Data Generation, Autonomous Reasoning

KEY PROJECTS

SPENTRA-SSM- Physics-Guided Underwater Image Enhancement Framework (Spectral-Prior Physics Enhanced Neuron Texture Restoration Architecture) <i>Physics-Guided Restoration · VMamba · PyTorch · GANs · OpenCV</i> - Developed a physics-guided underwater restoration framework integrating NAFNet, VMamba State Space Models (SSM), spectral chroma synthesis, and adaptive fusion architectures. - Designed cross-phase semantic fusion with attention-driven feature injection, structural consistency refinement, and frequency-domain enhancement for underwater semantic realism. - Implemented spectral consistency learning, memory fusion mechanisms, explainable feature-flow visualization, and adversarial optimization pipelines in PyTorch. - Optimized restoration using SSIM, MS-SSIM, perceptual, spectral reconstruction, and adversarial loss functions for robust underwater enhancement performance.	- DEC 2025 – MAY 2026
SPENTRA-GEN - Synthetic Underwater Data Generation Engine <i>Synthetic Data Generation · Physics Simulation · PyTorch · OpenCV</i> - Developed a physics-aware synthetic underwater data generation framework with controllable optical degradation simulation and supervision-map generation. - Implemented underwater physics modeling: scattering, attenuation, turbidity, wavelength absorption, backscatter synthesis, and caustic distortion simulation. - Generated auxiliary supervision maps (depth, transmission, segmentation, scattering priors) for multi-task underwater AI training pipelines. Designed configurable rendering and augmentation workflows to improve domain generalization and downstream underwater vision performance.	- JULY 2025 – NOV 2025
SRRE-V3 - Self-Referential Reasoning Engine <i>Agentic AI · FastAPI · Llama 3 · LLM Orchestration</i> - Architected an autonomous reasoning framework for resource-constrained LLMs using iterative self-correction, adversarial validation, and multi-perspective reasoning pipelines. - Developed closed-loop cognitive modules for structural logic repair, debate-driven strategy evaluation, hybrid reasoning synthesis, and dynamic rejection gating. - Engineered memory-aware reasoning orchestration with constraint-enforced generation and self-reflective critique systems using Llama 3 via Ollama and FastAPI inference pipelines. - Designed modular AI-agent coordination enabling adaptive reasoning depth, recursive validation, and hallucination-aware response filtering.	- JAN 2025 – MAY 2025

Graph-Enhanced Transformer (GET) - Reducing Hallucinations in LLMs

- AUG 2024 – NOV 2024

Transformers · Knowledge Graphs · DistilBERT · NLP · Generative AI

- Developed a Graph-Enhanced Transformer integrating Knowledge Graphs into transformer architectures to reduce hallucinations in Large Language Models.
- Designed a KG-Augmented Attention Mechanism modifying transformer self-attention to prioritize factual relationships using knowledge graph embeddings.
- Built a prototype using DistilBERT, PyTorch, and Hugging Face Transformers integrated with Wikidata-based knowledge graphs for training and evaluation.
- Improved factual consistency and contextual grounding through graph-aware semantic attention and structured knowledge injection.

ECG-AI - Deep Learning Framework for Cardiovascular Signal Intelligence

- JUNE 2023 – MAY 2024

PyTorch · AlexNet · VGG16 · Healthcare AI · Signal Processing

- Developed a deep learning ECG classification framework for automated detection of Atrial Fibrillation (AF), Normal Sinus Rhythm (NSR), and Congestive Heart Failure (CHF) using MIT-BIH datasets.
- Engineered advanced signal preprocessing integrating FFT, Butterworth filtering, wavelet transforms, denoising, and 1D-to-2D ECG spectral transformation for enhanced feature representation.
- Evaluated Custom CNN, VGG16, and AlexNet architectures, achieving 99.21% classification accuracy with high precision, recall, and F1-score for cardiac abnormality detection.
- Optimized model generalization through augmentation, normalization, spectral feature extraction, and cross-validation-based performance analysis.

Advanced Multilingual Sentiment Analysis & Hate Speech Detection

- DEC 2022 – APRIL 2023

NLP · BERT · RoBERTa · PyTorch · Deep Learning

- Developed a multilingual NLP platform for detecting sentiment polarity and hate speech across multiple languages using BERT and RoBERTa transformer architectures.
- Engineered a deep learning inference pipeline achieving 92% accuracy in detecting complex semantic and linguistic variations across cross-lingual datasets.
- Implemented preprocessing, tokenization, contextual embedding extraction, and multilingual dataset balancing for robust NLP performance.

INTERNSHIPS

Artificial Intelligence Internship

May 2023 – Sep 2023

MathWorks Supported

Data Analytics Internship

Dec 2022 – Feb 2023

AWS Academy Supported

AI-ML Internship

Mar 2022 – May 2022

AWS Academy Supported

Web Development Internship

Jun 2020 – Aug 2020

Internshala Supported

CERTIFICATIONS

MathWorks - Deep Learning, Image Processing, Machine Learning

AWS - Data Analytics, Machine Learning

IIT Bombay - Python 3.4.3 Training, Java Training Certification

IBM - Python for Data Science

HackerRank - Problem Solving, Python, SQL